Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*AI-Driven Active Transfer Learning for Efficient Process Modeling in Advanced Manufacturing.*				
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (✓), Next-gen Communication ( ), Advanced Robotics & Manufacturing (✓), Quantum ( )				
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)			**Technology Field 2** (Large Category)	
	Mechanical ・ Materials			*Artificial Intelligence*	
	Medium Category		Small Category	Medium Category	Small Category
	Additive Manufacturing/ Bio 3DPrinting		*Digital manufacturing*	*Smart Manufacturing*	*AI-Based Manufacturing Process Control*
Concept of Proposed Technology	The proposed technology aims to develop an Active Transfer Learning framework for data-efficient manufacturing process modeling that enables accurate prediction and optimization of manufacturing outcomes under new materials, machines, or process conditions with minimal experimental effort. Manufacturing companies often possess large volumes of historical process data collected under previous conditions, while data collection under new conditions is expensive, time-consuming, and resource intensive. Direct reuse of such legacy data, however, frequently leads to negative transfer, where differences between historical and new operating conditions or related but non-identical domains				

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	degrade model performance rather than improve it. To address this challenge, the proposed technology introduces an adaptive learning system that actively controls both knowledge transfer and data acquisition. The framework integrates transfer learning with active sampling to selectively reuse only the most informative source data while actively querying new experiments in the target domain when additional information is truly needed. A key feature of the approach is the explicit modeling and mitigation of negative transfer risk, ensuring that transferred knowledge improves, rather than harms, predictive accuracy. The technology leverages probabilistic AI models with uncertainty quantification to guide decision-making, enabling the system to balance the cost of new experiments against the expected benefit of reusing prior data. By jointly optimizing source data selection and target sampling, the proposed approach significantly reduces the number of required experiments while maintaining robust and reliable process models. This technology is broadly applicable to advanced manufacturing domains, including additive manufacturing, biomanufacturing, composites, and materials processing, and provides a scalable foundation for smart manufacturing systems that accelerate process development, reduce cost, and improve model reliability across varying production conditions. **Key features:** • Negative transfer–aware knowledge reuse for selectively incorporating informative legacy data from related manufacturing domains while mitigating performance degradation • Joint source selection and active target sampling to balance the cost of new experiments with the benefit of transferred knowledge • Uncertainty-guided process modeling for reliable prediction and decision-making under limited, noisy, and expensive experimental data
Rationale for Joint R&D Project	The proposed project requires close collaboration between our research team at Georgia Tech and an industrial partner to successfully translate active transfer learning methodologies into a practical manufacturing process modeling and decision-support system. A joint R&D framework is essential to integrate advanced AI algorithms with real manufacturing data, experimental workflows, and operational constraints, and to validate the technology under realistic industrial conditions. **Rationale for collaboration:** • The Georgia Tech team provides expertise in active learning, transfer learning, uncertainty-aware modeling, and negative transfer mitigation for manufacturing process modeling. • The Korean SME partner provides domain knowledge, manufacturing process data, and experimental or production environments required for realistic validation. • Effective co-design of AI algorithms and manufacturing workflows is necessary to balance model accuracy, experimental cost, and robustness across varying process conditions. **Expected benefits of joint R&D:** • Accelerated transition from AI research to deployable manufacturing process modeling tools.

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	• Reduced technical and experimental risk through iterative algorithm process co-development and validation. • Creation of a scalable foundation for commercialization of data-efficient, AI-enabled manufacturing solutions applicable across materials and process variants.
Key Words	1. AI-Enabled Smart Manufacturing 2. Active Transfer Learning 3. Data-Efficient Manufacturing Process Modeling 4. Negative Transfer Mitigation 5. Adaptive Sampling and Experimentation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop an active transfer learning-based manufacturing process modeling system that can reliably predict and optimize manufacturing outcomes under new materials, machines, and operating conditions by jointly integrating knowledge transfer and adaptive data acquisition through advanced artificial intelligence. The specific R&D objectives are: • To develop negative transfer-aware transfer learning algorithms that selectively reuse informative data from related but non-identical manufacturing domains while preventing performance degradation • To develop adaptive source selection and active target sampling strategies that balance experimental cost and model accuracy for efficient manufacturing process modeling • To develop uncertainty-aware AI models for robust manufacturing process prediction and decision support under limited, noisy, and expensive experimental data			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Experimental Data Reduction:** Reduce the number of required target-domain experiments by at least 40% while maintaining comparable or improved predictive performance relative to conventional modeling approaches without active transfer learning. **Negative Transfer Mitigation:** Ensure that the proposed framework avoids negative transfer in more than 90% of evaluated scenarios, as measured by maintaining target model performance equal to or better than target-only learning. **Model Prediction Accuracy Improvement:** Achieve at least a 30% reduction in prediction error (e.g., RMSE or MAE) on target manufacturing processes compared to target-only models using the same number of experimental samples. **Adaptive Sampling Efficiency:** Achieve at least a 25% improvement in sampling efficiency, measured by faster convergence to a predefined prediction accuracy threshold compared to passive or one-shot transfer learning strategies.			

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R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy that advances from algorithmic development and simulation-based studies to laboratory-scale validation and finally to pilot-scale industrial demonstration in realistic manufacturing environments.
	
	• **Year 1: Algorithm Development and Simulation-Based Validation**
	  ○ Develop the core active transfer learning framework for manufacturing process modeling, integrating transfer learning, uncertainty-aware modeling, and adaptive sampling
	  ○ Implement negative transfer–aware source selection strategies, including forward selection and backward elimination methods
	  ○ Develop transfer-aware acquisition functions for guiding adaptive target-domain sampling under limited data
	  ○ Validate core algorithms using synthetic benchmarks and simulation-based manufacturing case studies, assessing prediction accuracy, negative transfer mitigation, and sampling efficiency
	
	• **Year 2: Laboratory-Scale Integration and Manufacturing Process Validation**
	  ○ Integrate the active transfer learning framework into a prototype manufacturing process modeling and decision-support system
	  ○ Apply the developed methods to laboratory-scale manufacturing processes, incorporating real experimental data from selected materials, machines, or process conditions
	  ○ Customize and optimize algorithms to account for measurement noise, process variability, and experimental constraints
	  ○ Conduct repeated laboratory experiments to evaluate model robustness, data efficiency, and predictive performance compared to conventional modeling approaches
	
	• **Year 3: Pilot-Scale Demonstration and Technology Advancement**
	  ○ Deploy the integrated system in a pilot-scale industrial or SME-relevant manufacturing environment
	  ○ Perform large-scale validation studies across multiple materials, machines, or operating conditions to verify generalizability and robustness
	  ○ Refine system performance, usability, and reliability based on experimental feedback and industrial requirements
	  ○ Assess commercialization potential and prepare the technology for industrial transfer, including documentation, workflow integration guidelines, and collaboration with SME partners

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Georgia Tech team will be responsible for the development of core AI algorithms and system-level intelligence for the proposed active transfer learning-based manufacturing process modeling technology. Georgia Tech team will lead the research on transfer learning, active sampling, uncertainty-aware modeling, and negative transfer mitigation required to achieve reliable and data-efficient manufacturing process prediction under new operating conditions. The specific roles and responsibilities of Georgia Tech team include: • Development of active transfer learning algorithms, including negative transfer-aware source selection, adaptive target sampling, and uncertainty-guided decision-making • Design of the overall system architecture and integration framework for AI-enabled manufacturing process modeling and decision support • Simulation and laboratory based development, algorithm validation, and performance analysis using synthetic benchmarks and manufacturing-relevant case studies Technical guidance and close collaboration with the industrial partner to support system integration, experimental validation, and pilot-scale demonstration of the proposed technology.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the application, integration, and validation of the proposed active transfer learning technology within real manufacturing workflows and environments. The SME will play a key role in translating AI algorithms into a practical, deployable manufacturing process modeling and decision-support system aligned with industrial constraints and operational needs. The specific roles and responsibilities of the Korean SME partner include: • Provision of manufacturing process data, experimental facilities, and domain expertise across selected materials, machines, or operating conditions relevant to the project • Integration of AI algorithms into existing manufacturing data pipelines, experimental workflows, or production systems • Execution of laboratory-scale and pilot-scale experiments, including data collection, testing, and performance evaluation under realistic manufacturing conditions • Contribution to technology refinement, robustness improvement, and usability enhancement, leveraging industrial experience and operational feedback • Participation in commercialization planning and technology transfer activities, including identification of use cases, scalability considerations, and deployment pathways Close collaboration with the Georgia Tech team to ensure effective co-development, validation, and industrial relevance of the proposed technology.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver a prototype active transfer learning-based manufacturing process modeling and decision-support system capable of accurately predicting and optimizing manufacturing outcomes under new materials, machines, or operating conditions with significantly reduced experimental effort. The project will advance the state of the art in data efficient manufacturing intelligence by integrating negative transfer-aware knowledge reuse with adaptive sampling strategies, achieving measurable improvements in prediction accuracy, robustness, and experimental efficiency. From an industrial perspective, this project will provide the Korean SME partner with core AI-enabled manufacturing modeling technologies and practical integration know-how that can be applied across a range of manufacturing processes, including additive manufacturing, biomanufacturing, and materials processing. The outcomes will support technology commercialization and product development, enabling SMEs to reduce development cost, accelerate process optimization, and improve decision-making under limited data conditions. More broadly, the project is expected to strengthen the competitiveness of Korean manufacturing SMEs, promote adoption of AI-driven smart manufacturing solutions, and contribute to the growth of the advanced manufacturing and industrial AI ecosystem through sustained international collaboration.

6. References

1. R. B. Gramacy. *Surrogates*. Chapman and Hall/CRC, mar 2020. ISBN 9780367815493. doi: 10.1201/9780367815493. URL https://www.taylorfrancis.com/books/9781000766202
2. Bracco, F., Zanderigo, G., Paynabar, K., & Colosimo, B. M. (2025). Leveraging transfer learning for efficient bioprinting. Biofabrication.
3. Tang, Y., Rahmani Dehaghani, M., Sajadi, P., & Wang, G. G. (2024). Selecting subsets of source data for transfer learning with applications in metal additive manufacturing. Journal of Intelligent Manufacturing, 1-22.
4. Aboutaleb, A. M., Bian, L., Elwany, A., Shamsaei, N., Thompson, S. M., & Tapia, G. (2017). Accelerated process optimization for laser-based additive manufacturing by leveraging similar prior studies. IISE Transactions, 49(1), 31-44.
5. Cheng, L., Tsung, F., & Wang, A. (2017). A statistical transfer learning perspective for modeling shape deviations in additive manufacturing. IEEE Robotics and Automation Letters, 2(4), 1988-1993.
6. Ren, J., & Wang, H. (2019). Surface variation modeling by fusing multiresolution spatially nonstationary data under a transfer learning framework. Journal of Manufacturing Science and Engineering, 141(1), 011002.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization